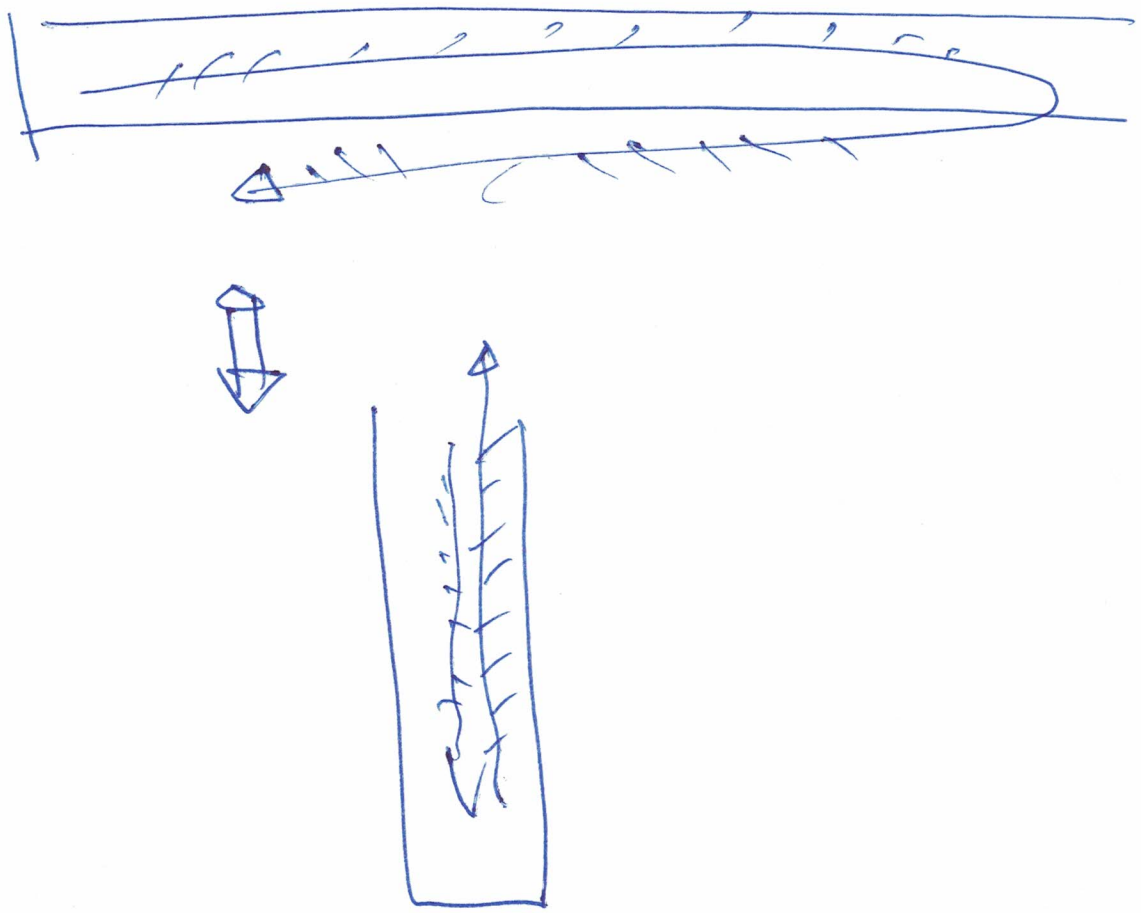
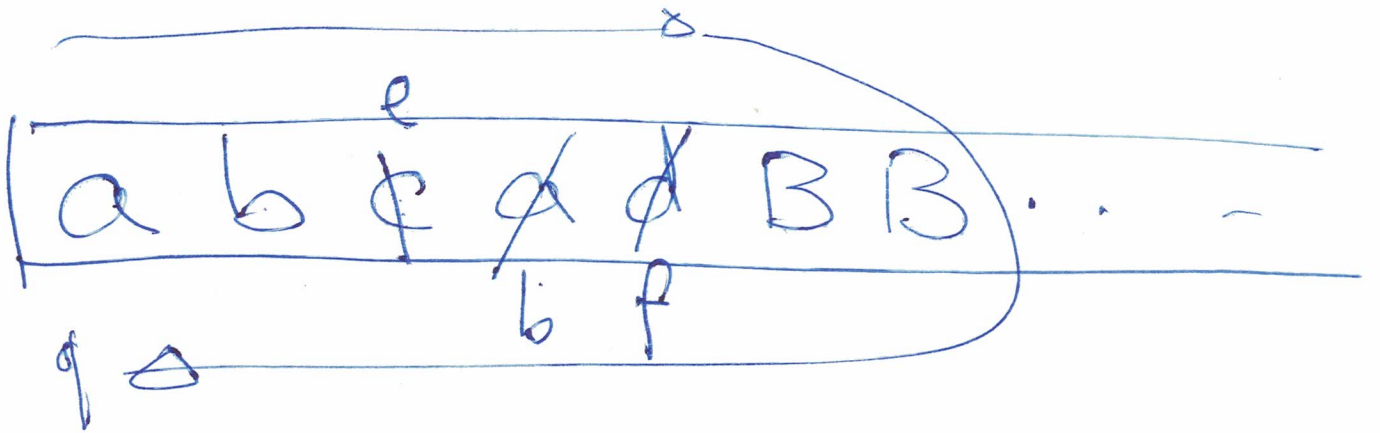
Cpts 317. 11/28

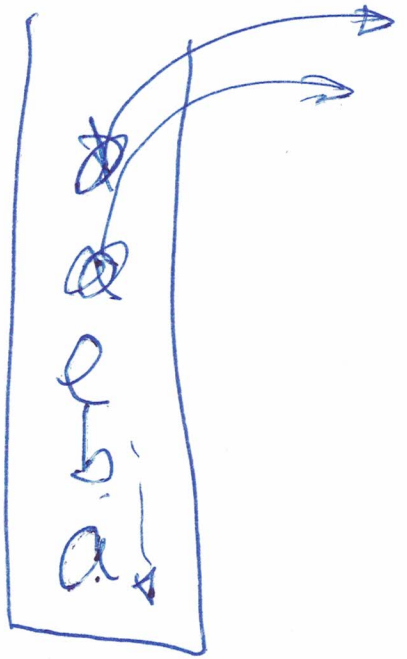
HW Hint:

① & ②: standard.

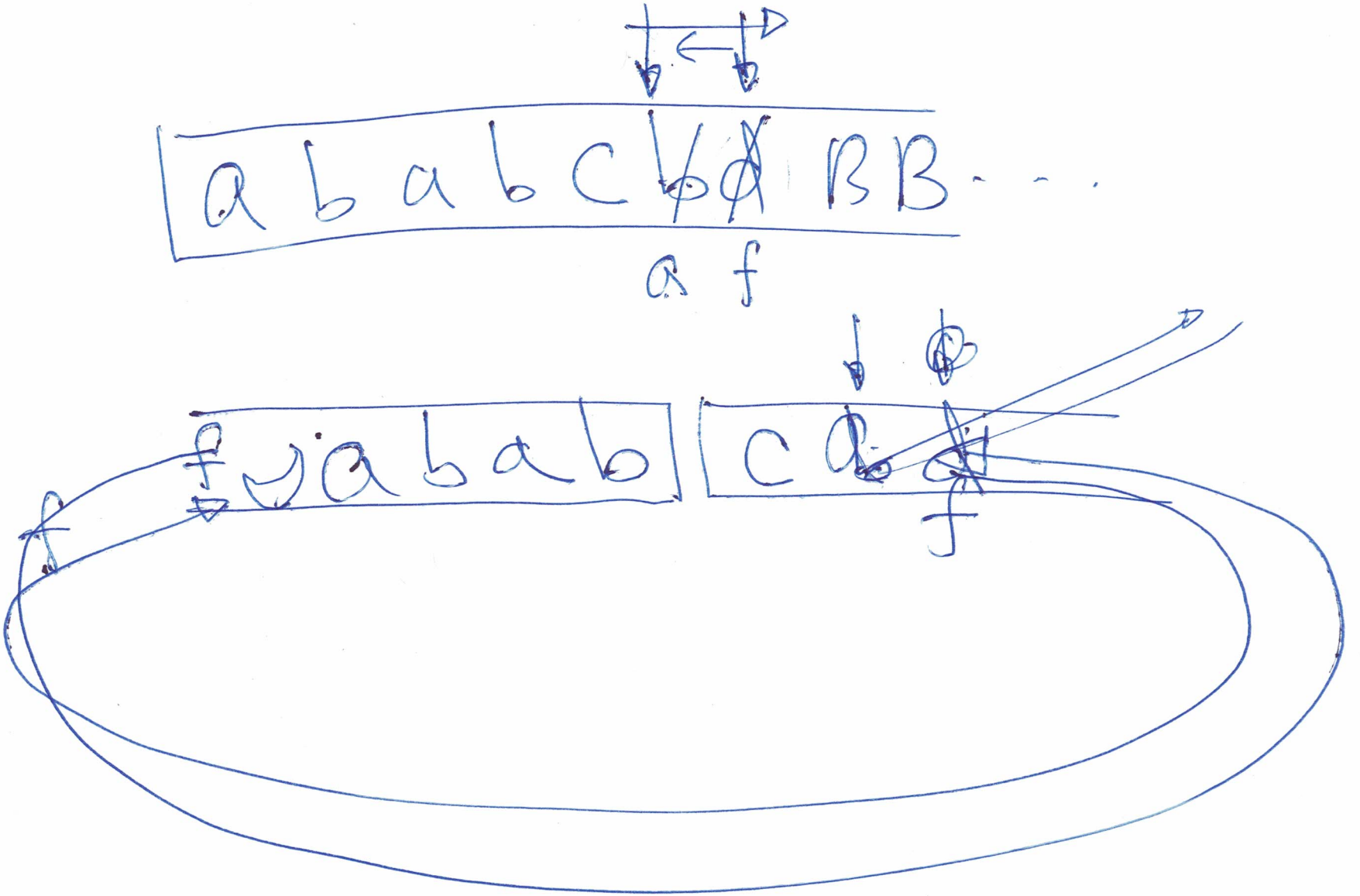
③:

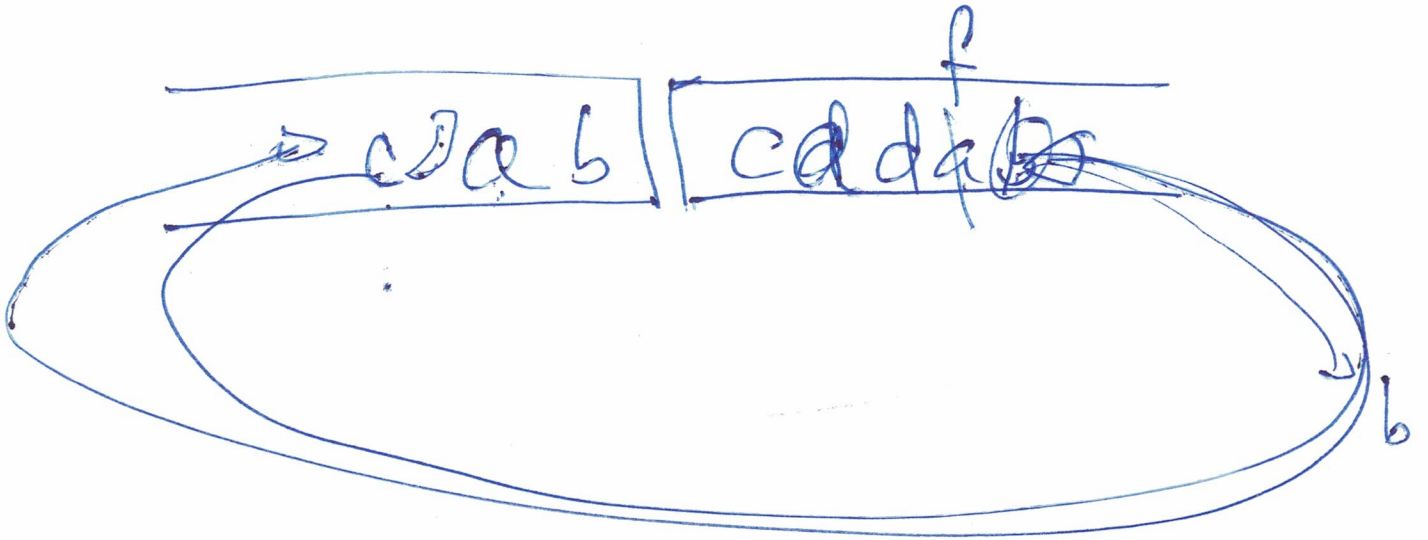




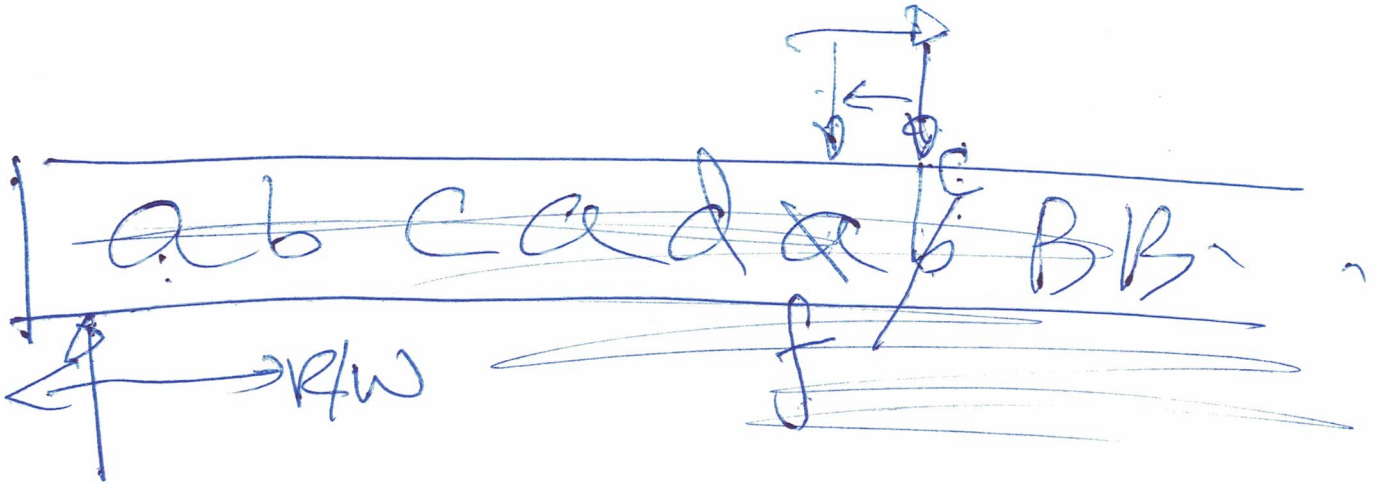


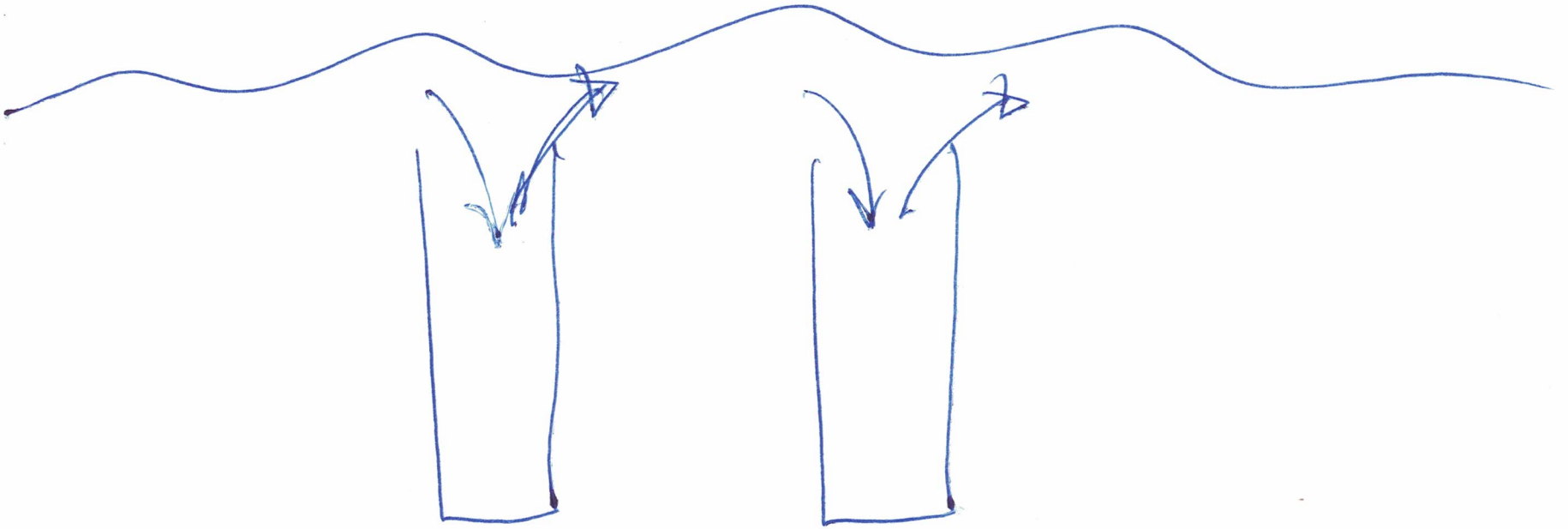
Keep "updated" tape content in the stack when head moves from left-to-right.


a b a b c ¢ ¢ B B . . .
a f
f ababab c a ¢
f
f


f
b

④. TM tape ≡ two stacks.





Recall: L is r.e. if $L = L(M)$ for some TM M;

That is, $\forall x$,

$x \in L$ iff M says yes on x.

L is recursive if $L = L(M)$ for some ~~TM~~ algorithm M

(M halts on all input; L is recog. by M):

That is, $\forall x$

$x \in L$ iff M says yes on x;

$x \notin L$ iff M says no on x.

Show that: if L_1 and L_2 are r.e., then so is $L_1 \cup L_2$.

Proof. Since L_1 and L_2 are r.e., so we have TMs M_1 and M_2 to accept them, resp.

To show $L_1 \cup L_2$ is r.e., we need only build a TM M to accept it:

```
M: input x
   Run M1 on x;
   if M1 says yes,
      ret yes;
   else Run M2 on x;
      if M2 says yes,
         ret yes.
```

?

Clearly, M accepts $L_1 \cup L_2$ ∎

Wrong proof!

Proof. (corrected). I construct M to accept $L_1 \cup L_2$ as follows:

M: input x

Run M_1 and M_2 in parallel on x;

if one of the two says yes,

ret yes.

do it:

Thm. if L_1 and L_2 are r.e.,
Then so is $L_1 \cap L_2$.

Thm. if L_1 and L_2 are recursive,
Then so is $L_1 \cup L_2$.
(Then, so is $L_1 \cap L_2$)

More proofs.

Thm. Let L be a recursive language.

Define $L' = \{x : \exists y.\ xy \in L\}$.

Show that L' is r.e.

Proof. Since L is recursive, we have an alg M (halting on all input) to recog. L: $\forall x$,

$x \in L \Rightarrow M$ says yes on x;

$x \notin L \Rightarrow M$ says no on x.

To show that L' is r.e., we need construct a TM M' to accept it: $\forall x$,

$x \in L' \Rightarrow M'$ says yes on x;

$x \notin L' \Rightarrow M'$ doesn't say yes on x;

```
M':   input x
      int n=1;        <- inf. many
      // Enumerate all the y's from short to long
      //
      while(1){
        for each word y with length n
            Run M on xy
            if M says yes        // xy ∈ L
                ret yes
        n++;
      }
```

Clearly, M' accepts L'.

do it:

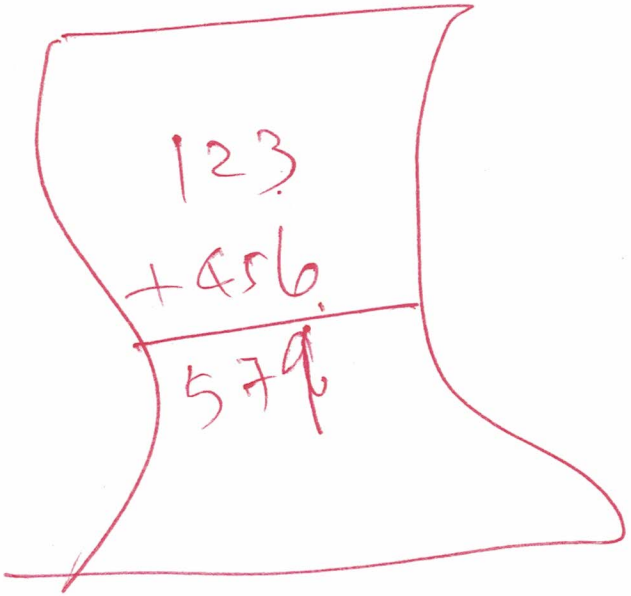
Thm. if L is recursive, then

So is $L' = \{x : \exists y.\ xy \in L$ and $|x| = |y|\}$.

Summary: ①. TMs and C-programs are equivalent;

②. TMs are powerful; Turing-Church Thesis: all computers built by humans (past, present, future) can't be more powerful than TMs;

③.



global read & write?

Next: Thm. there is a language that is r.e. but not recursive.